

$$d) \lim_{x \rightarrow -\infty} \left(\frac{3x+2}{3x-1} \right)^{2x} = \left(\frac{x \cdot (3 + \frac{2}{x})}{x \cdot (3 - \frac{1}{x})} \right)^{2x} = \frac{\lim_{x \rightarrow -\infty} 3 \cdot (1 + \frac{2}{3x})}{\lim_{x \rightarrow -\infty} 3 \cdot (1 - \frac{1}{3x})} = \frac{\lim_{x \rightarrow -\infty} \left[1 + \frac{2}{x} \right]^{\frac{2}{3}}}{\lim_{x \rightarrow -\infty} \left[1 + \frac{1}{x} \right]^{\frac{2}{3}}} = \frac{e^{\frac{4}{3}}}{e^{\frac{2}{3}}} = e$$

$$\frac{1}{x} = \frac{2}{3x} \quad x = \frac{2x}{3} \quad \frac{1}{m} = \frac{1}{3x} \quad x = -\frac{m}{3}$$

$$⑤ \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \ln(e)$$

$$a) \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{x} = \lim_{x \rightarrow 0} \frac{e^u - 1}{\frac{u}{2}} = 2 \ln e = 2, \quad 2x = u \quad x = \frac{u}{2}$$

$$b) \lim_{x \rightarrow 0} \frac{2^{3x} - 1}{x} = \lim_{x \rightarrow 0} \frac{2^u - 1}{\frac{u}{3}} = 3 \ln 2 \quad 3x = u \quad x = \frac{u}{3}$$

$$c) \lim_{x \rightarrow 0} \frac{3^{2x} - 1}{\frac{e^{5x \ln 2} - 1}{e^{5x \ln 2} - 1}} = \frac{(e^{2x \ln 3} - 1)}{(e^{5x \ln 2} - 1)} \cdot \frac{2x \ln 3}{5x \ln 2} = \frac{(e^{2x \ln 3} - 1)}{(e^{5x \ln 2} - 1)} \cdot \frac{2 \ln 3}{5 \ln 2} = \frac{\ln e}{\ln e} \cdot \frac{2 \ln 3}{5 \ln 2} = \frac{2 \ln 3}{5 \ln 2}$$

$$2^{5x} - 1 = (e^{5x \ln 2} - 1)$$

$$3^{2x} - 1 = (e^{2x \ln 3} - 1)$$

$$d) \lim_{x \rightarrow 0} \frac{e^{2x} - 1}{e^{3x} - 1} = \frac{e^{2x} - 1}{e^{3x} - 1} \cdot \frac{\frac{2x}{e^{2x} - 1}}{\frac{3x}{e^{3x} - 1}} = \frac{e^{2x} - 1}{e^{3x} - 1} \cdot \frac{2x}{3x} = \frac{\ln e}{\ln e} \cdot \frac{2}{3} = \frac{2}{3},$$

$$e) \lim_{x \rightarrow 2} \frac{e^x - e^2}{x - 2} = \frac{e^x - e^2}{x - 2} = \frac{e^{m+2} - e^2}{m} = \frac{e^m \cdot e^2 - e^2}{m} = \frac{e^2 (e^m - 1)}{m} = e^2,$$

$$m = x - 2 \quad x = m + 2$$

* A forma indeterminada $\frac{0}{0}$ aparece quando $0 = 2$

$$f) \lim_{x \rightarrow 2} \frac{2^x - 2^0}{x - 0} \rightarrow \frac{2^x - 4}{x - 2} = \frac{e^{x \ln 2} - e^{2 \ln 2}}{x - 2} = \frac{e^{2 \ln 2} (e^{(x-2) \ln 2} - 1)}{x - 2} = \frac{4 \cdot (e^{(x-2) \ln 2} - 1)}{x - 2} = \lim_{x \rightarrow 0} \frac{4 \cdot (e^x - 1)}{\frac{x}{\ln 2}} = \lim_{x \rightarrow 0} 4 \ln 2 \cdot \left(\frac{e^x - 1}{x} \right) = 4 \ln 2,$$

$$x = (x-2) \ln 2 \quad (x-2) = \frac{x}{\ln 2} \quad x-2 \rightarrow 0 \quad x \rightarrow 0$$

$$g) \lim_{x \rightarrow 0} \frac{e^x - e^0}{x - 0} = \lim_{x \rightarrow 0} \frac{e^x - e^0}{x} = \frac{e^x \cdot e^0 - e^0}{x} = \frac{e^0 (e^x - 1)}{x} = e^0,$$

$$x = x - 0$$

$$x = x + 0$$